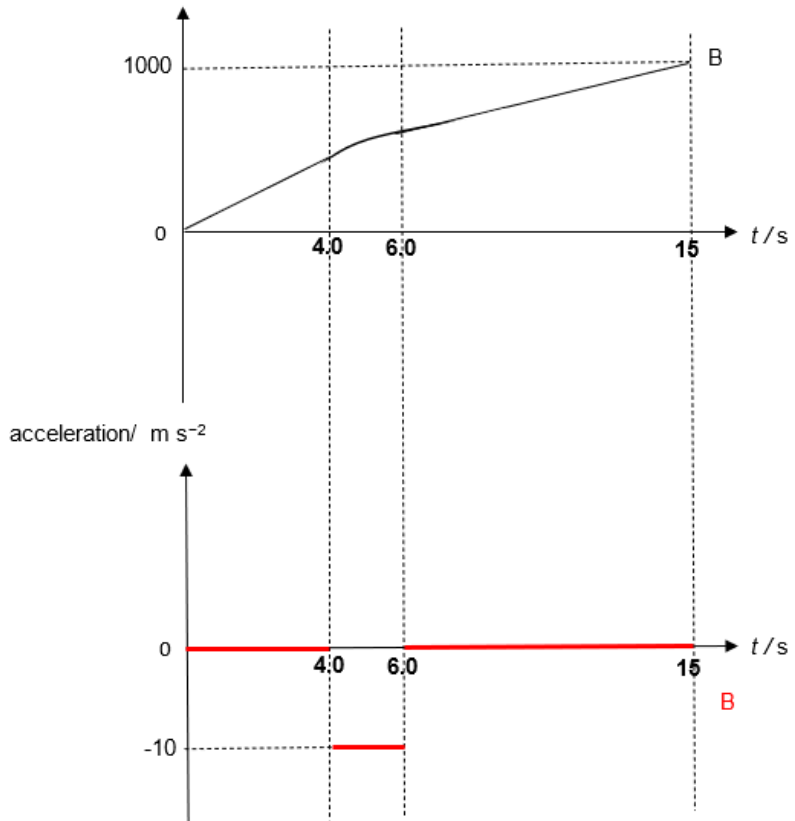
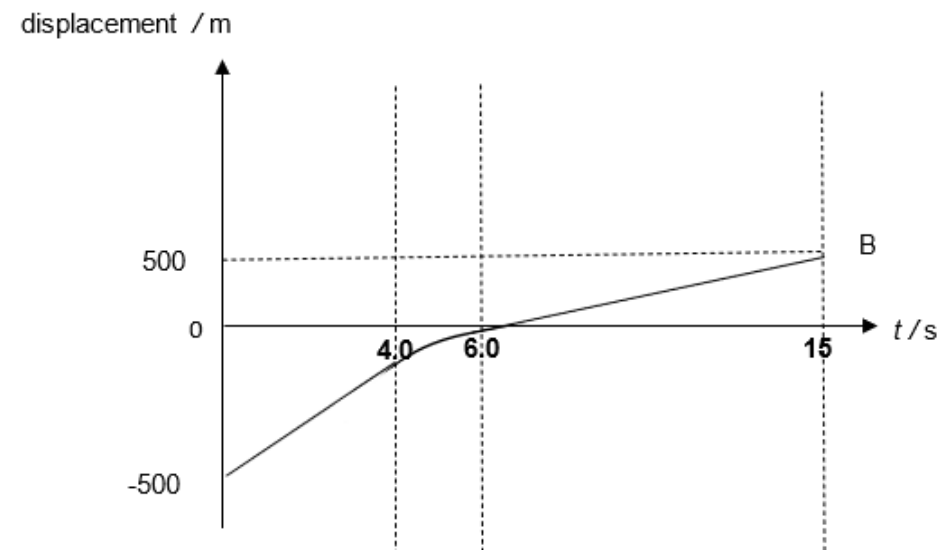


Qns	Answer	Marks
1(a)(i)	Precision is determined by the range in the values. Range of Set A = $1.55 - 1.43 = 0.12 \text{ cm}$ Range of Set B = $1.58 - 1.42 = 0.16 \text{ cm}$ Range of Set C = $1.56 - 1.45 = 0.11 \text{ cm}$ Therefore, Set C is more precise as the range of values are smaller than the other two sets.	M1 for all correct values A1
1(a)(ii)	Accuracy is determined by the closeness of the measured value to the true value. Average values for Set A = $\frac{1.52 + 1.49 + 1.43 + 1.52 + 1.55}{5} = 1.502 = 1.50$ Average values for Set B = $\frac{1.42 + 1.45 + 1.43 + 1.51 + 1.58}{5} = 1.478 = 1.48$ Average values for Set C = $\frac{1.45 + 1.56 + 1.47 + 1.53 + 1.46}{5} = 1.494 = 1.49$ Therefore, Set A is more accurate as the average values are closer to the true value of the diameter.	M1 for all correct values A1

Qns	Answer	Marks
1(b)(i)	displacement / m  acceleration/ m s^{-2} <u>Alternative s-t graph</u>  <u>For s-t graph:</u> 1m – for correct sketch of graph for car B and correct displacement value Also, gradient of the line for first 4.0s must be greater than the gradient of the line for the last 9.0s <u>For a-t graph:</u> 1m – for correct sketch of graph for car B and correct acceleration value	B1 B1

Qns	Answer	Marks
1(b)(ii)	Minimum distance occurs when both cars have the same speed ($v = 60 \text{ m s}^{-1}$). Distance covered by Car A during the 6.0 s = $\left[\frac{1}{2} \times (25 + 60) \times 6.0 \right] = 255 \text{ m}$ Distance covered by Car B during the 6.0 s : $(80 \times 4.0) + \left[\frac{1}{2} \times (80 + 60) \times 2.0 \right] = 460 \text{ m}$ Minimum distance = $(500 + 255) - 460 = 295 \text{ m}$	B1 C1 for both correct answer A1

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Qns	Answer	Marks
2(c)	No resultant force (in any direction)	D1